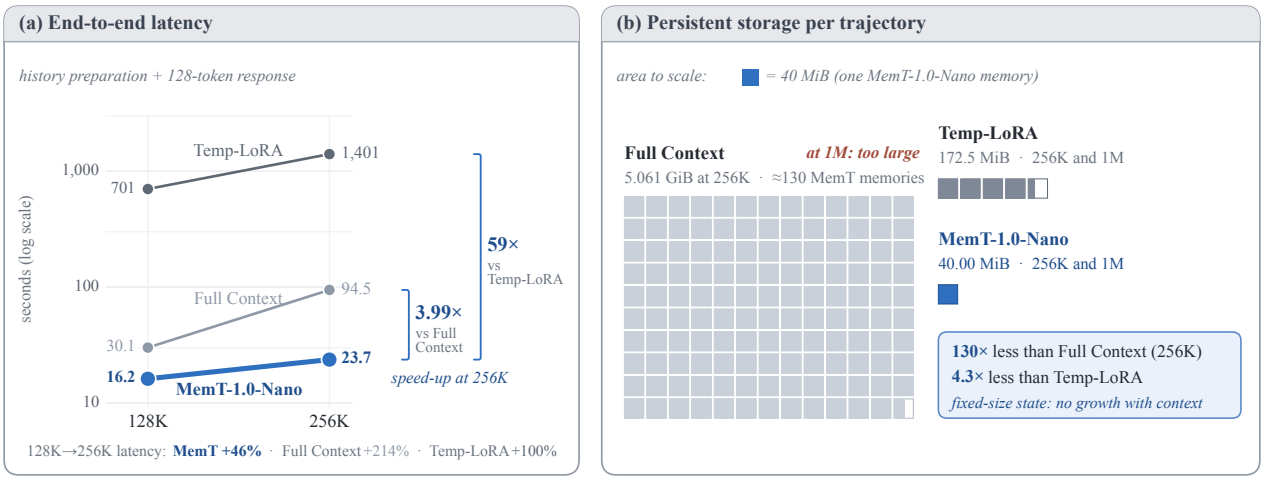

(a) End-to-end latency
history preparation + 128-token response
seconds (log scale)
1,000
100
10
Temp-LoRA
1,401
701
Full Context
94.5
30.1
23.7
16.2
MemT-1.0-Nano
128K
256K
59×
vs
Temp-LoRA
3.99×
vs Full
Context
speed-up at 256K
128K→256K latency: MemT +46% · Full Context +214% · Temp-LoRA +100%
(b) Persistent storage per trajectory
area to scale:
= 40 MiB (one MemT-1.0-Nano memory)
Full Context
at 1M: too large
5.061 GiB at 256K · ≈130 MemT memories
Temp-LoRA
172.5 MiB · 256K and 1M
MemT-1.0-Nano
40.00 MiB · 256K and 1M
130× less than Full Context (256K)
4.3× less than Temp-LoRA
fixed-size state: no growth with context